

ANALYSTLAB AFRICA
INTERNSHIP EXPERIENCE - WEEK 1 PROJECT
FINTRUST DIGITAL BANK

A. Business Problem Statement

1. What problem is FinTrust trying to solve?

FinTrust has customer and transaction data, but the information is not yet being used in a structured way to understand customer behaviour, transaction failures, and financial risk. The main problem is turning the raw data into useful information that management can use to identify issues and improve digital banking services.

2. Why is the problem important?

The problem is important because poor understanding of transaction and customer data can make it difficult to detect risky transactions, reduce failed transactions, understand valuable customer groups, and know which service channels are performing well. Having reliable information can help FinTrust make better decisions about its customers and digital services.

3. How can the Data Analyst track contribute to solving it?

As a Data Analyst Intern, I can contribute by cleaning and profiling the data, checking its quality, analyzing transaction and customer patterns, and developing KPIs that show important changes in the business. I can then present the findings through Power BI or Tableau so that management can easily identify areas that need attention.

4. What type of output could your track provide?

My track could provide a data quality report, KPI scorecard, exploratory analysis, and a dashboard showing transaction trends, customer segments, transaction failures, risk-review patterns, and channel performance. I could also provide a wireframe showing how these findings could be organized into a final Power BI or Tableau dashboard.

B. Resource Review & Profiling Methodology

Resource Name	Contents and Coverage	Expected Usage in Analysis	Identified Limitations
FinTrust Customer Data.csv	Customer ID, Customer name, age, gender, city, customer segment, account type, tenure months, digital engagement, monthly income, preferred channel, account status	Identify customer segmentation and demographic profiling	Dataset is synthetic and may require validation for completeness

FinTrust Transaction Data.csv	Transaction ID, Customer ID, Transaction DateTime, Transaction Type, Amount NGN, Channel, Device Type, Location, International Transaction, Transaction Status, Risk Review Flag	I would use this dataset to analyze transaction patterns, customer behavior, channel usage, and transaction performance across time, location, and device types. I would also examine risk-review patterns by comparing flagged and non-flagged transactions based on transaction amount, type, channel, international status, and transaction outcome.	The dataset is synthetic, so its transaction and risk patterns may not fully reflect real-world customer behavior and require validation before drawing practical conclusions.
FinTrust Data Dictionary.xlsx	Field name, Dataset, Data type, Example, Definitions, Business use, Modelling use	I would use this dataset to understand the structure and meaning of different fields, assess how each variable can support business decisions, and identify which variables could be useful for analytical or machine-learning models.	The dataset provides metadata rather than actual observations, so it cannot be used to measure trends, relationships, or performance directly.

C. Track objectives

1. To assess the quality and structure of the customer and transaction data by checking for missing values, duplicates, inconsistent formats, and other data-quality issues.
2. To analyze customer and transaction patterns to understand transaction behaviour, customer activity, transaction types, channels, and other important characteristics.
3. To develop relevant KPIs and exploratory insights that can help FinTrust monitor transaction performance, customer activity, transaction failures, and risk-review patterns.
4. To identify useful customer and transaction patterns that may help management better understand high-value customers, service-channel usage, and areas requiring further investigation.

5. To design a clear dashboard structure in Power BI or Tableau that presents the key findings in a way that management can easily understand and use.

D. Identifying Success Criteria

I would consider my work successful if I can produce a clean and well-documented dataset, identify meaningful patterns from the data, and develop KPIs that accurately represent the areas being analyzed. I would also check whether the dashboard presents the findings clearly, whether the calculations can be reproduced from the source data, and whether the final analysis gives management useful information for further investigation and decision-making.

C. Developing initial plan

Week No	Task	Key Focus	Milestone	Measurable Completion	Expected Output
2	Data preparation and profiling	Review customer and transaction datasets; understand fields and data types; check missing values, duplicates, and inconsistencies; begin basic exploration.	Complete the initial data profiling and document the main data-quality issues.	• Profile 100% of available fields • Document the data type and purpose of each field • Identify and record missing values, duplicates, and inconsistencies	Data profiling report, data-quality findings, prepared working dataset, initial observations
3	Data analysis and KPI Development	Analyze customer activity, transaction trends, failures, risk-review patterns, and channel performance; develop relevant KPIs; create initial visualizations.	Complete the main exploratory analysis and establish the KPIs for the dashboard.	• Answer at least 4 business questions • Develop at least 5 KPIs • Create at least 5 supporting visualizations • Document the calculation/definition of each KPI • Identify at least 3 meaningful findings	Exploratory analysis, KPI definitions, visualizations, key findings
4	Dashboard Development and review	Build the Power BI/Tableau dashboard; incorporate KPIs and findings; check calculations and visual clarity;	Complete and review a working dashboard containing the agreed KPIs and	• Include at least 5 validated KPIs • Include at least 5 relevant visualizations • Check 100%	Working dashboard, KPI scorecard, key findings, limitations, final

		document findings and limitations.	analysis	of displayed KPI calculations against the source data • Document at least 3 key findings • Document at least 2 data limitations • Present the dashboard for feedback	presentation
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Data Analytics Track

A-Business Understanding and Data Review

1. What business questions should FinTrust's management be able to answer?

As a Data Analyst Intern the questions that FinTrust's management be able to answer should be: how many transactions are being made, what is the total value over time, which transaction types and channels are used most frequently by customers, which customer groups generate high transaction value, which patterns are associated with transactions that receive a risk-review flag, are international transactions showing different patterns from domestic transactions?

2. What decisions could data analysis support?

The analysis could support management in several areas including risk management, transaction reliability, understanding customer activity and identify different patterns of customer usage, service-channel decisions, use transaction trends to understand periods of higher activity and plan resources accordingly, and provide KPIs that allow management to track transaction activity and failures.

3. What stakeholders could benefit from the analytical outputs?

Stakeholders could use the outputs to understand how customers use channels and devices and also monitor overall transaction performance and customer activity.

C. Analytical Questions

1. What transaction types are most frequently used by customers, and how does transaction activity vary across these types?
2. What is the total and average transaction value across different transaction types?
3. Which transaction channels are used most frequently, and what proportion of transactions does each channel account for?
4. Which channels have the highest transaction failure rates?
5. How does transaction activity and transaction value change over time?
6. What proportion of transactions are successful, failed, or have other transaction statuses, and how does this vary by transaction type or channel?.

7. What characteristics are common among transactions that receive a Risk Review Flag, particularly in terms of transaction amount, channel, device type, location, transaction type, and international status?
8. Are international transactions associated with different transaction values, channels, or risk-review patterns compared with domestic transactions?

D. KPI Planning

KPI Name	Business Definition	Why it matters	Required Data Fields
Total Transaction	- The total number of transactions processed within a specified period. - Total Transaction = Count of Transaction ID	It provides a measure of the overall level of transaction activity and helps management monitor changes in transaction volume over time and across different channels or transaction types.	Transaction ID
Transaction success rate (TSR)	- Percentage of total transactions completed successfully versus failed or reversed. - TSR = Number of Successful Transactions / Total number of transactions * 100	Measures operational stability and reliability of technical infrastructure.	Transaction_ID, Status,
Average Transaction Value (ATV)	- Total monetary value of transactions divided by the total count of transactions. - ATV = Sum of Amount of NGN / Count of Transaction ID	Indicates spending behavior and customer account usage density.	Transaction_Amount, Transaction_ID
Channel Transaction Share (CTS)	- Share transactions processed per channel. - Number of transaction through a specified channel / total transactions * 100	Guides infrastructure investment and resource allocation per channel	Channel, Transaction_ID

Risk Flag Transaction Value (RFTV)	- Number of transactions marked with Risk_Review_Flag as a percentage of total number of transactions. $RFTV \% = \frac{\text{Total amount for RFTV}}{\text{Total amount for all transactions}} * 100$	Quantifies potential financial risk exposure needing manuel/compliance review.	Transaction_Amount, Risk_Review_Flag , Status
International Transaction Rate (ITR)	- The percentage of total transactions that are international transactions within a specified period. $\text{International Transaction Rate (\%)} = \frac{\text{Number of International Transactions}}{\text{Total number of Transactions}} * 100$	It helps management understand the level of international transaction activity and compare international activity with domestic transactions. It can also support further analysis of transaction value, channels, and risk-review patterns.	International Transaction, Transaction ID

E. Dashboard Planning and Wireframe

- [*Proposed FinTrust Digital Bank Transaction Dashboard Wireframe*](#)

F. Initial Analysis Plan

The analysis will be conducted using Excel to transform the raw FinTrust transaction data into useful business insights. The process will begin with data validation and profiling, followed by exploratory analysis, KPI calculation, and visualization. The analysis will focus on transaction activity, transaction value, transaction status, channel performance, customer transaction behaviour, and risk-review patterns.

1. Data Preparation and Validation

- Review the transaction dataset and confirm the fields and data types.
- Check for missing values, duplicate Transaction IDs, inconsistent categories, and invalid values.
- Check the Transaction DateTime field for consistency and ensure transaction amounts are numeric.

- Standardize categorical values where necessary to support accurate analysis.

2. Exploratory Analysis

- Analyze transaction volume and value across transaction types.
- Examine transaction activity across different channels.
- Analyze transaction activity over time.
- Compare successful, failed, and other transaction statuses.
- Compare international and domestic transactions.
- Examine transaction activity across available device types and locations.

3. Risk Review Analysis

- Analyze the number of risk-flagged transactions by channel.
- Compare risk-flagged transactions across transaction types.
- Compare risk-review patterns between international and domestic transactions.
- Examine risk-flagged transactions by device type.
- Compare the total transaction value of risk-flagged and non-flagged transactions.

4. KPI Calculation

The following KPIs will be calculated from the transaction data:

- Total Transactions
- Total Transaction Value
- Average Transaction Value
- Transaction Success Rate
- Risk-Flagged Transaction Value %
- International Transaction Rate

The KPI calculations will be validated against the underlying transaction data to ensure accuracy.

5. Visualization

PivotTables and Excel charts will be used to present the analysis clearly. The planned visualizations include:

- Transaction Volume — Line Chart
- Transactions by Type — Bar Chart
- Transactions by Channel — Bar Chart
- Transaction Status — Donut Chart
- Transaction Value by Channel — Bar Chart
- Risk-Flagged Transactions by Channel — Bar Chart
- Risk Review by International Transaction Status — Column Chart
- Risk-Flagged Transactions by Transaction Type — Bar Chart
- Risk-Flagged Transactions by value — Column Chart

6. Expected Analytical Output

The analysis is expected to provide a clearer understanding of transaction behaviour, transaction performance, channel usage, transaction reliability, transaction value patterns, and risk-review activity. The findings will be used to identify areas that may require further investigation and to support the development of the FinTrust management dashboard.

G. Future Dependencies

- Clean and validated transaction data, required for accurate KPI calculations and analysis.
- Complete data fields and consistent data definitions needed to ensure the analysis is interpreted correctly.
- Further data quality checks required before using the data for final reporting.
- Stakeholder feedback on KPIs and dashboard design needed to confirm that the selected metrics and visuals meet management requirements.
- Access to Power BI, if required later may be needed for development of the final interactive dashboard.
- Additional customer-level data, if provided could support deeper customer segmentation and behavioural analysis.